DIGITAL SIGNATURE

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Abstract

There are different types of encryption techniques are being used to ensure the privacy of data transmitted over internet. Digital Signature is a mathematical scheme which ensures the privacy of conversation, integrity of data, authenticity of digital message/sender and non-repudiation of sender. Digital Signature is embedded in some hardware device or also exits as a file on a storage device. Digital Signature are signed by third party some certifying authority. This paper describe the different key factor of digital signature with the working of digital signature, through various methods and procedures involved in signing the data or message by using digital signature. It introduces algorithms used in digital signatures.

Keywords: *Encryption, Hashing, Public Key Encryption, Authentication, Privacy, Information Security*

I. INTRODUCTION

With the revolution of digital data storage and communication technologies, digital information can be easily stored, copied, changed and transported. These desirable properties are very useful as well as having some security issues; therefore digital is regarded as unreliable in areas where privacy, authentication and integrity of data are of big concern unless some security procedures attached to it. There are some areas like receipt, contracts, and similar others where users have concerns of unauthorized access, and stealing of data. Conventional signatures are not able to change this situation because they are included in the document as a part of the document.

The solution of all these security issues is Digital Signature. When we sign a document digitally, we send the signature as a separate document. For a Digital Signature, the recipient receives the message and the signature. The recipient needs to apply a verification technique to the combination of the message and the signature to verify the authenticity. Digital Signature ensure the privacy if data and prevent it from unauthorized access.

Digital Signature is currently used in various application domains that include: i) Government: Filing tax returns online by taxpayers, citizen ID card, issuing forms and licenses, reservations & ticketing, ii) Banking: Inter/ Intra bank messaging systems, corporate Internet banking applications, iii) Financial Services/Broking: Online trading, electronic contract notes, iv) B2B: Online tendering, e-Procurement, v) Healthcare: healthcare management system, electronic medical recording, electronic prescriptions and many others.

This paper is organized as follows: section 2 briefly describes privacy, authentication, integrity and non-repudiation. Section 3 introduces encryption and in section 4 illustrates steps involved in Digital Signatures.

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II. KEY FACTORS OF DIGITAL SIGNATURE [1]

Privacy, Authentication, Integrity and Non-repudiation are four key factors to achieve information security. Privacy also called confidentiality, guarantees protection of information from unauthorized persons.

A. Privacy

Privacy guarantees the security of data from unauthorized access and manipulation of data. It means that a transaction between businesses cannot be viewed or interfered with by third party.

B. Authentication

Authentication serves as proof that you are the real person not the third party intruder. Authentication is very critical in case of any trust between different parties. Authentication is critical if there is to be any trust between parties. Authentication is also needed when user communicate over the network and login to network in order to prevent data from alteration.

C. Integrity

Integrity refers to the ability to protect information, data, or transmissions from unauthorized, uncontrolled, or accidental alterations. We can also use the term integrity in reference to the working/functioning of system, network and application. By preventing unauthorized or undesirable changes to data, data integrity can be achieved, which ensures the internal and external consistency and other data attributes such as accuracy, completeness etc.

D. Non-Repudiation

Non-Repudiation provides the security against denial by third unauthorized party which involved in the communication or having participation in the communication. Thus when the message is send to the receiver, then receiver can prove that the alleged sender in fact sent the message.

III. ENCRYPTION

Encryption is the conversion of the data into a form, called a cipher text that cannot be easily understood by unauthorized people. Decryption is the process of converting encrypted data back into its original form, so it can be understood.

For recovering the contents of the encrypted message, we require correct decryption key. Decryption key is an algorithm which decrypts the encrypted message. If the encryption algorithm is more complex then more difficult it becomes to eavesdrop on the communications without access to the key. Any computed system which implements one or more specific algorithms is called Cryptosystem.

A. Symmetric encryption

Symmetric encryption or private key encryption is a form of cryptosystem in which encryption and decryption are performed by using the same key. It is also known as the conventional encryption. It transforms plaintext into cipher text using a secret key and an encryption algorithm. Traditional symmetric ciphers use substitution and /or transposition techniques. Substitution techniques map plain text elements into cipher text elements. Transposition techniques systematically replace the positions of plaintext elements.

An algorithm that implements Secret Key Encryption is called Symmetric Algorithm. Classic symmetric encryption [3] can be achieved using following techniques: Caesar, Vigenère, Hill, Monoalphabetic substitution, Playfair, ADFGVX, Byte Addition, Exclusive-OR, Vernam,

homophonic substitution, permutation, and Solitaire. Modern Symmetric Encryption [3] can be achieved using following techniques: IDEA, RC2, RC4, DES in ECB mode, and DES in CBC mode, Triple-DES in ECB mode and Triple-DES in CBC mode, Rijndael, and AES. Secret Key Encryption techniques require secure and sophisticated mechanism to securely distribute key to all parties.

B. Asymmetric encryption

Asymmetric encryption is a form of cryptosystem in which encryption and decryption are performed using the different keys-one a public key and one private key. This is also known as public key encryption. It transforms plaintext into cipher text using a one of two keys and encryption algorithm. Symmetric encryption offers confidentiality, authentication or both. RSA is the most widely used public key cryptosystem.

Some Asymmetric Encryption algorithms are: RSA, DSA, and ECDSA. It offers key advantages like simplified key distribution, Digital Signature and long-term encryption. Public key authentication is often used when authentication should be performed automatically without user intervention [4].

IV. DIGITAL SIGNATURE

Digital Signature is an authentication mechanism that enables the sender of a message to attach a unique code that act as a signature typically the signature is formed by taking the hash of the message and encrypting the message with the senders private key. The signature guarantees the source and integrity of the message. The Digital Signature standard is an NIST standard that uses the secure hash algorithm. The plain message, the message signature and the Public Key of the sender are packed together which is transformed into signed and encrypted message using the Public Key of the recipient. The recipient unpacks received message which is the signed and encrypted message after which same hashing function is used to compute message digest of the received message which is compared to the decrypted signature.

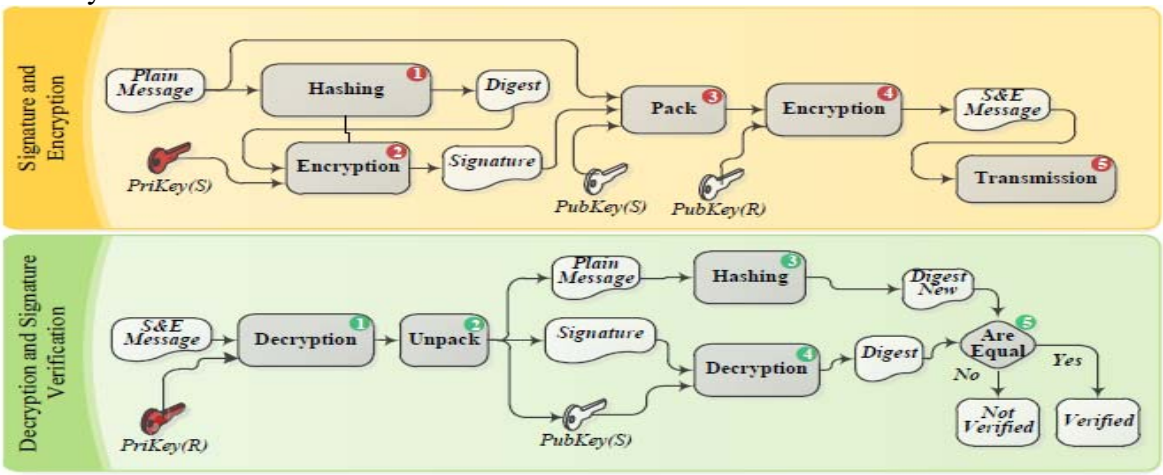

Figure1. Functioning of Digital Signature

Digital Signature can be classified into two processes:

1. Signing and encryption
2. Decryption and verification

Steps followed in these processes are describes as follow:

Process 1: Signing and Encryption

1 Hashing: In this step small message digest is computed which is unique representation of the message. This evaluation ensures the message integrity. The digital signature is applied to this smaller message digest. This evaluation generates a unique code.

2 Encryption: In this step message digest is encrypted using private key of the sender. It is used to sign the message digest. The original message can be recovered by decrypting the message signature using corresponding public key of sender. To obtain non-repudiation, Signing is performed.

3 Packing: The plain message, message signature and the Public Key of the sender are packed together into a single packed unit.

4 Encryption: The single packed unit of message which contains plain message and message signature along with the public of the sender is encrypted using receipt's public key to form signed and encrypted message.

Process2: Decryption and Verification

1 Decryption: In these steps the received message which is signed and encrypted is decrypted using the receiver's private key to form a packed unit of message containing plain message, the signature and the public key of the sender.

2 Unpack: The decrypted message in last step is unpacked into plain text message, message signature and the public key of the sender.

3 Hashing: In this Step, plain text message which is obtained after decrypting and unpacking the received message is input to hash function that was used by sender is used to compute message digest.

4 Decryption: In this step, the received message signature is decrypted by using the received public key of the sender. By decryption, message digest is obtained computed before the transmitting the message.

5 Comparison: At last in this step, message digest obtained after decrypting the received message signature and message digest computed from the plain message received by the recipient.

Signed and encrypted data or message can only be decrypted using the correct Private Key of the recipient thus ensuring the privacy. Hashing and Signature verification offer the integrity, authenticity and non-repudiation.

A. Digital Signature (using symmetric key)

Encryption and decryption algorithms using asymmetric keys are too slow to be used for long messages, thus a symmetric key is generated and is used in Digital Signatures to encrypt the packed unit containing plain message, the message signature and Public Key of sender. The Symmetric Key is encrypted using the Public Key of the recipient so that it can only be decoded by the intended receiver who can use it to decrypt the packed unit before unpacking it and drawing semantics. The functioning of signing and verification is illustrated in figure 2.

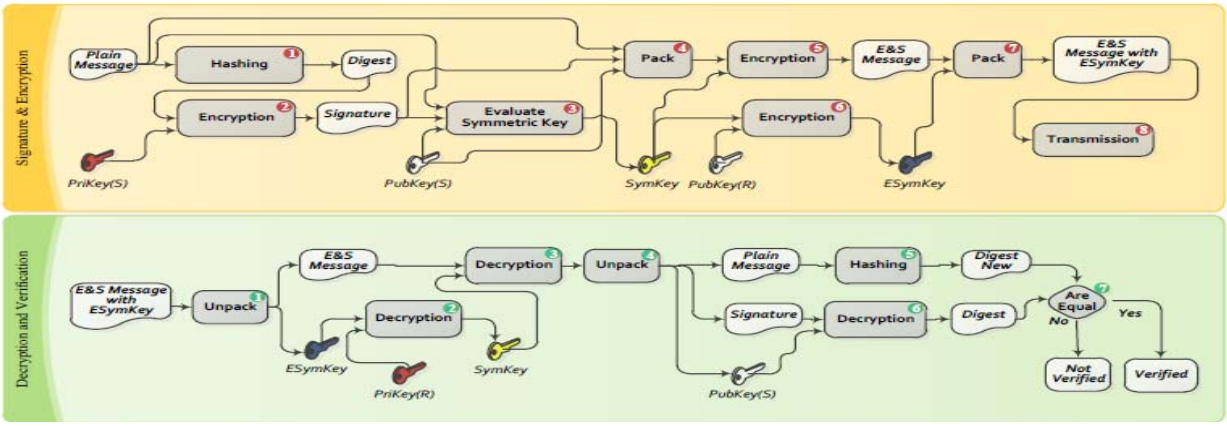

Figure2. Functioning of Digital Signature (using Symmetric Key)

Steps followed in each process are describes as follow:

Process1: Signing and Encryption

1 Hashing: This step is same as that for Digital Signature detailed in signed and encryption step 1 of section 4.

2 Encryption: This step is same as that for Digital Signature detailed in signed and encryption step 2 of section 4.

3 Symmetric Key Evaluation: A Symmetric Key is calculated using some algorithm based on plain message, message signature and the public key of the sender. Any algorithm may be used for calculating the Symmetric Key.

4 Packing: The plain message, message signature and the Public Key of the sender are packed together to form a single packed unit.

5 Encryption: The packed message containing the plain message and the signature of the message in the form of encrypted digest along with the Public Key of the sender is encrypted using the Symmetric Key calculated in step 3 to form signed and encrypted message.

6 Encryption: The Symmetric Key is encrypted to form encrypted Symmetric Key using Public Key of the recipient

7 Packing: The encrypted and signed message and encrypted Symmetric Key are packed into a single unit called encrypted and signed message with encrypted symmetric key.

Process: Decryption and Verification

1 Unpack: The received encrypted and signed message with encrypted symmetric key is unpacked in this step.

2 Decryption: The encrypted symmetric key is decrypted using the Private Key of the recipient to get the Symmetric Key used for encryption in step 5 of signing and encryption.

3 Decryption: Decryption: In this step the received message which is signed and encrypted is decrypted using the Symmetric Key decrypted in step 2 to form a packed message containing plain message, the signature and the public key of the sender.

4 Unpack: This step is same as for Digital Signature detailed in step 2 of decryption and verification of section 4.

5 Hashing: This step is same as for Digital Signature detailed in step 3 of decryption and verification of section 4.

6 Decrypt: This step is same as for Digital Signature detailed in step 4 of decryption and verification of section 4.

7 Comparison: This step is same as for Digital Signature detailed in step 5 of decryption and verification of section 4.

The process of this scheme is similar to that of Digital Signature without use of Symmetric Key and offers privacy, integrity, authentication and non- repudiation and also offers additional advantage of faster encryption and decryption. However, addional algorithm is still required to evaluate symmetric key.

V. ALGORITHMS USED IN DIGITAL SIGNATURE

A. Hash Algorithms

Hash function H maps a variable length message M as input and produce a fixed sized hash values.

$$h = H(M) \quad (1)$$

Various hash algorithms, their characteristics and hash size are listed in table 1.

Table 1: Hash Algorithms

Name of Algorithm	**Type and Characteristics**	Hash Size
Secure Hash Algorithm 1 (SHA1) [8]	FIPS approved; other versions (SHA256, SHA384, SHA512) provide longer outputs	160 bits
Message Digest 5(MD5) [9]	Potential weaknesses is that it can be used as a keyed hash	128bits
RACE Integrity Primitives Evaluation Message Digest 160 (RIPEMD-160) [10]	Developed as part of the EC's Research and Development in Advanced Communications Technologies in Europe (RACE)	160 bit
TIGER Hash [11]	Designed for efficient operation on 64-bit platforms	192 bits

B. Digital Signature Algorithm

Digital Signature Algorithm (DSA), Rivest-Shamir-Adleman (RSA) , and Elliptic Curve DSA (ECDSA) which are captured in FIPS PUB 186-2 (with change notice 1 dated 5 October 2001) are three Digital Signature algorithms. These algorithms used for digital signature along with their characteristics and minimum key size are listed in table 2

Table 2: Digital Signature Algorithms

Name of Algorithm	**Type and Characteristics**	**Min. Key Size**
Digital Signature Standard (DSS) [5]	FIPS 186-2 digital signature Digital signature based on SHA1 hash, unencumbered (no patents, no licenses)	1024bits
RSA Digital Signature [6]	RSA digital signature (FIPS approved) Previously patented digital signature	1024 bits
Elliptic Curve Digital Signature (ECDSA) [7]	Digital signature based on elliptic curve key technology uses smaller keys than other public key technologies but may be encumbered by various	160 bits

VI. CONCLUSION

Various encryption techniques are being used to ensure privacy and authentication of digital information. Digital Signatures employs encryption, hashing and Digital Signature algorithms to ease its users attain all four desired properties privacy, integrity, authentication and non-repudiation for information security. There are several possible ways to use Digital Signatures each having its pros and cons.

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